

Programmable In-Network Timing Obfuscation for DNP3

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Abstract—Device fingerprinting is an early step in reconnaissance against industrial control systems. On a power-grid control network, a passive observer can tell what kind of operation a DNP3 outstation is performing from the time the device itself takes to answer. Existing traffic obfuscation does not transfer directly to this setting for two reasons. The features differ: fingerprinting of control devices uses the behaviour of the physical device, such as the interval between the transport acknowledgment and the application response, rather than the flow-level sizes and inter-packet times that general traffic obfuscation reshapes. The assumptions differ too: techniques that mix obfuscated with normal traffic rely on an encrypted channel to keep the two apart, while these networks still carry plaintext because of the legacy devices on them, so padding or dummy packets are visible as such. A defense here therefore has to change timing without changing bytes, and without modifying a protective relay whose firmware is fixed.

We present an in-network timing-obfuscation framework that runs in the data plane of one programmable switch placed in front of the outstation. The framework classifies each DNP3 exchange, holds the packets that carry the outstation's timing, and releases them on operator-set offsets, so the interval an observer records is a policy value instead of the device's own. The intervention is timing only: no packet changes size. We implement it as a P4 program on an Intel Tofino-1 and evaluate it against a physical SEL-751A relay over 22 grouped collection runs in one approximately five-hour campaign, comprising 63,360 DNP3 exchanges.

A measured sweep of the release policy on the switch shows that the interval an observer records follows the configured offset across a usable range while the end-to-end response time stays fixed, and that the range is closed from above by the finite fail-open horizon. Under the operating policy the cross-layer response time, the feature this class of reconnaissance relies on, is replaced by the policy value: its median moves to the configured release and its interquartile range falls from about 2.8 ms to 0.006 ms. For the evaluated fixed Random-Forest attacker, one that trains on Timing OFF transaction-class timing before the defence is deployed and then applies that model unchanged, three-class transaction identification falls from 0.651 balanced accuracy to approximately three-class chance, and the mutual information between the interval and the transaction class falls from 0.383 bits to a value inside a within-run permutation null. An attacker that instead retrains on obfuscated traffic recovers part of the signal from the difference between the read-path and control-path anchors, which bounds what the mechanism achieves. On the master-facing link the framework adds no frame and no byte for the same workload, and costs about 21 to 23 ms

of added response time per exchange, with no retransmission and no timeout observed across 63,360 exchanges. These results characterise one relay behind one switch over one campaign. The plaintext DNP3 function codes remain visible and already name each operation, so what the mechanism removes is the additional, device-derived timing signature of that operation rather than the operation itself. A fixed release budget leaves a small tail outside the target interval. The classifier we evaluate separates transaction classes on a single outstation, so this is the suppression of a class-conditioned timing feature and not a demonstration of resistance to device identification.

Index Terms—traffic obfuscation, device fingerprinting, DNP3, programmable switches.

I. INTRODUCTION

Device fingerprinting has been an essential step in cyber reconnaissance, allowing adversaries to reveal unique features of target networks and to design effective, stealthy attack strategies. These techniques are becoming increasingly critical in industrial control systems (ICSs) such as power grids, where adversaries often use IP-based control networks and computing devices within as entry points to inflict physical damage. Consequently, fingerprinting shifts the focus from visited websites and user biometric behavior to device models and types of control operations that are critical to ICS attacks. In the 2015 attack that disrupted Ukrainian power grids and the Stuxnet attack that disrupted Iranian nuclear power facilities, it is widely believed that adversaries stay in their systems for at least 6 months to perform cyber reconnaissance [1], [2].

To disrupt device fingerprinting, many studies present network traffic obfuscation [3], [4]. Device fingerprinting targeting general computing environments generally relies on network-level features such as packet size and/or inter-packet latency observed from communication patterns [5]. Consequently, existing traffic obfuscation often focuses on (i) padding and splitting network packets, which hide or change the distribution of network packet sizes [3], and (ii) delaying network packets and adding dummy ones, which disrupt the inter-packet timing pattern [4], [6]. Since manipulating communication networks can introduce runtime overhead, recent works have begun to offload traffic obfuscation onto programmable network switches, which change communication patterns at much higher line rates than CPUs [7]–[9].

Unfortunately, these methods cannot be directly applied to ICS environments for two main reasons. First, device fingerprinting methods on ICS devices rely on different features from the ones in general computing environments. Instead of using network layer knowledge, they attempt to use behavior on physical devices to reveal device model or types

of operations. For example, work in [10] uses the latency between the TCP acknowledge packets and the actual response from the same device to estimate execution time there. Since each vendors may choose different materials or algorithms to implement the same control functionality, this time can be accurate in fingerprinting those devices. Second, existing traffic obfuscation is performed over encrypted channels, which are necessary to mix the obfuscated and normal traffic. Unfortunately, ICS networks still rely on unencrypted traffic due to a wide range of legacy devices, despite ICS network protocols may provide security features. Directly padding or adding dummy packets can easily reveal obfuscated packets, exposing the the underlying traffic pattern to adversaries.

In this paper, we present a framework that schedules when selected packets of an ICS transaction leave the network, so that a timing feature an observer measures becomes a value we choose rather than a property of the device. A programmable switch at the outstation edge holds the acknowledgment and the response of a transaction and releases each at a deadline it computes when the transaction begins. Because the switch forwards the same packets it received, the endpoints run unmodified firmware and every DNP3 byte reaches the master unchanged, which is what the plaintext setting requires. The framework is general over which packets are held and what the released interval should be. We demonstrate it on the two timing features that Formby et al. [10] use, the cross-layer response time of a poll and the master-visible timing of a control command, and we evaluate the first of these in depth. These are two case studies of one mechanism, not two systems.

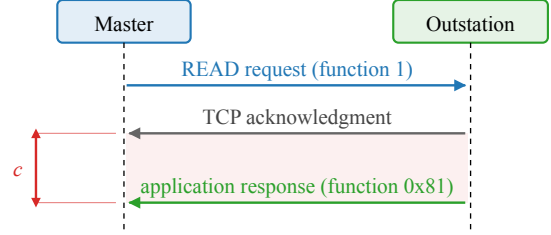
This paper makes the following contributions:

- **Design a timing-obfuscation framework** that replaces a selected within-transaction timing feature with a configured value by scheduling the release of real protocol packets, and derive the coupling between the acknowledgment hold, the response hold, and the target interval that any such schedule must satisfy (Section IV).
- **Realize the framework on an Intel Tofino-1** using paired blocker and hold queues that sustain a hold until a deadline expires, with an acknowledgment-anchored read lane, a request-anchored control lane, and forwarding that stays open when a response arrives after its scheduled release (Section V).
- **Evaluate the realization against a physical SEL-751A relay** and report how closely the released interval reaches its target, what the policy can and cannot cover, the latency the mechanism adds, the hardware resources it uses, and what an adaptive attacker still learns from the features that remain (Section VI).

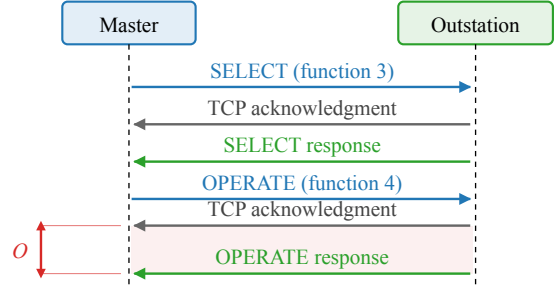
II. BACKGROUND AND MOTIVATION

In this section we focus on the DNP3 transactions the paper covers, the features that leak from them and how an adversary uses them in fingerprinting, and the opportunity that a programmable data plane gives us to change the timing without touching the endpoints.

(a) READ poll



(b) Select-before-operate control



c : acknowledgment-to-response interval of the READ (CLRT).
 O : master-visible OPERATE response-to-ACK interval.

Fig. 1. The DNP3 transactions the paper measures, as seen on the master-facing link; requests in blue, transport-layer acknowledgments in grey, and application responses in green. (a) A READ poll: the request, the outstation's transport-layer acknowledgment, and its application-layer response. The interval c between the acknowledgment and the response is the cross-layer response time (CLRT). (b) A select-before-operate control: SELECT and OPERATE each produce an acknowledgment and a solicited application response. The interval O between the OPERATE's acknowledgment and its solicited response is the master-visible OPERATE response-to-ACK interval.

The DNP3 read and control transactions. DNP3 carries supervisory control and data acquisition (SCADA) traffic over TCP between a master and an outstation [11]. The master polls the outstation with a READ request (function code 1), and the outstation returns the requested measurements in an application-layer response (function code 0x81). A control command, on the other hand, uses select-before-operate (SBO): the master first sends a SELECT (function code 3) that names an output point, and the outstation returns a SELECT application response; the master then sends an OPERATE (function code 4), which the outstation confirms with a solicited OPERATE application response. A DNP3 control response repeats the control object, but it is a solicited application response, not a distinct message type. Because DNP3 runs over TCP, every request is also acknowledged at the transport layer before the application answers. As such, each transaction on the wire is a request, a TCP acknowledgment that carries no DNP3 payload, and a DNP3 response, in that order, as Figure 1 shows. We use these terms consistently in the rest of the paper: the acknowledgment is the TCP segment with the ACK flag that the outstation returns first, and the response is the DNP3 application frame that follows it.

Why the transaction timing is a fingerprint. The acknowl-

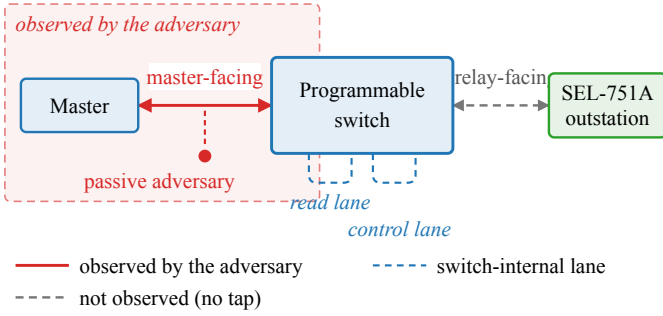


Fig. 2. Observation model. The master reaches the SEL-751A outstation through one programmable switch. The passive adversary sees only the master-facing link (solid red, shaded region); the relay-facing link and the two switch-internal scheduling lanes (dashed) are not observed, by the adversary or by our measurements.

edgment is produced by the outstation’s TCP stack as soon as the request arrives, whereas the response is produced by its application only after the device has done the work the request asks for. Consequently, the interval between the two, the CLRT, reflects the outstation’s internal processing rather than the network path, and it differs between device models and between the kinds of work they do. Formby et al. used it to tell device types and models apart in a substation network, and used the time a device takes to complete a control operation as a second feature that reveals the physical device behind the command [10]. Our own measurements on a physical SEL-751A relay show the same effect on one device: with no timing defense, the median CLRT is 2.116 ms for a READ, 2.050 ms for a SELECT and 2.937 ms for an OPERATE, each with a long tail, so the three transaction classes are already partly separable from timing alone (Section VI). The function codes remain visible in the plaintext, so the classes themselves are not a secret; what the timing adds is a channel that identifies the device and its operation without any payload, and that survives encryption.

Measured and inferred events. Everything the paper measures is a packet timestamp on the master-facing link, i.e., the request, the acknowledgment, and the response. The switch-internal release times and any event on the relay-facing link are not observed; where the paper names them, they are inferred from the configuration. Physical breaker motion is never measured.

Programmable data planes. A programmable switch exposes a match-action pipeline, written in P4 [12], that parses headers, keeps per-packet state in registers, and places packets into output queues at line rate. Because such a pipeline can delay and release a packet without a host in the loop, it can hold an outstation’s acknowledgment and response and release them on a schedule that the operator sets, while changing no endpoint and no byte of the payload.

III. THREAT MODEL AND OBJECTIVES

Assumptions on the adversary. The adversary is a passive observer on the master-facing link and has the ability to record

TCP and IP headers, segment sizes, and arrival times, but neither modifies nor injects packets, as in the fingerprinting attack of Formby et al. [10]. Figure 2 shows the setup. Because DNP3 is usually deployed without transport security across shared substation networks [13], we assume the traffic here is unencrypted. Therefore, the adversary is able to read the exchanges directly, including the DNP3 function codes, and from them it derives the transaction timing of Section II and can train a classifier that separates transaction classes from timing features. The adversary cannot compromise the master or the outstation, control the switch, or read the switch’s internal state. It also does not observe the relay-facing link, because in our deployment that link is inside the switch and no tap exists, and it cannot see physical breaker motion. An adversary that actively interacts with the traffic, sends its own probes, or changes the network is out of scope; we stay with the passive observer, which is the setting that prior traffic obfuscation work targets.

Reconnaissance target. The adversary’s objective is to fingerprint outstation devices and obtain knowledge to prepare for a later attack. Because control-related attacks such as false data injection attacks work only when the attacker knows the target devices [14], reconnaissance is the enabling step to identify the devices and their weaknesses. The fingerprinting process can succeed without decoding the payloads because it is done using the timing features of the master-outstation exchanges. Following [10], the two timings in scope are the cross-layer response time (CLRT), i.e., the interval between the transport-layer acknowledgment and the application-layer response of a READ or of the SELECT phase of SBO, and the master-visible OPERATE response-to-ACK interval, i.e., the interval between an OPERATE’s acknowledgment and its solicited application response. The CLRT reflects the outstation’s internal processing, while the second reflects the physical device behind a control command. The adversary also sees the request-to-acknowledgment interval, which we report as measured.

What the evaluated attacker separates. Device fingerprinting is the application that motivates this work, and it is the setting Formby et al. study. What our evaluation measures is narrower and we keep the two apart throughout. With one outstation on the testbed there is no second device to confuse it with, so we do not evaluate device identification and make no claim about it. What we measure is whether the timing of an exchange still reveals *which class of transaction* it was, on this device. The mechanism therefore suppresses a class-conditioned timing feature; it does not demonstrate resistance to device identification. The plaintext DNP3 function code already names the operation to anyone reading the traffic, so what the timing adds is a second, device-derived signature of the same event, and that signature is what the framework removes.

Two attacker models. A defence that is only evaluated against a classifier retrained on its own output answers the wrong question, so we separate two adversaries. The *fixed* adversary learns the transaction classes from undefended

traffic and applies that model unchanged once the defence is deployed; this is the adversary the reconnaissance setting describes, because the profile is built before the defence is met. The *adaptive* adversary collects obfuscated traffic and retrain on it, which is a strictly stronger assumption and bounds what remains learnable. We report both.

Research objectives. Our objective is therefore to remove these features from the master-facing observing adversary. We state it as five properties that the evaluation answers one by one.

- **RO1 (read timing).** The visible CLRT of READ and of the SELECT phase of SBO is a configurable policy value rather than the outstation's own processing signature.
- **RO2 (control timing).** The master-visible OPERATE response-to-ACK interval stays at a policy value while the relay-facing hold is drawn from a configured codebook.
- **RO3 (timing leakage).** The timing features carry less information about the transaction class, and a fixed adversary that trained on undefended traffic is reduced to approximately three-class chance. This is a statement about a class-conditioned timing feature on one device, not about device identification.
- **RO4 (release policy and coverage).** The release budget, the range of policies the mechanism can realise, the fraction of traffic it can cover, and the residual it leaves are characterised rather than assumed.
- **RO5 (cost and protocol preservation).** The latency the mechanism costs and the frames and bytes it adds are measured, and the external DNP3 workload still completes: every request is answered and every control operation returns a success status.

What is out of scope. The framework does not hide the function codes, which stay visible in plaintext DNP3; RO3 is about the timing channel, not the command semantics. It does not change packet sizes, and no size-related claim is made in this paper. Additionally, it does not claim that two different devices become indistinguishable: the evaluation has one outstation, and what it shows is that this outstation's timing signature is replaced by a policy signature for the exchanges the budget can cover. The relay-facing link is not instrumented, so the per-transaction hold and the release toward the outstation are configured rather than observed.

IV. DESIGN

An observer on the link between the master and the outstation sees when each packet of a transaction arrives. Some of those arrival times are produced by the device itself and therefore carry information about it. Our aim is to make a selected one of these intervals a value we choose rather than a value the device produces. This section first states the timing model for a single transaction, then explains which quantities are free to choose and what constrains them, and finally treats the control transaction, whose anchor has to differ. The hardware realization is deferred to Section V; a reader should be able to follow this section without knowing anything about P4.

A. A timing model for one transaction

Consider one READ transaction, drawn in Figure 3. The master sends a request. The outstation answers it twice: first its transport layer acknowledges the request, and later its application layer returns the response with the data. We write t_A for the instant the acknowledgment reaches the switch and t_R for the instant the response reaches it. The switch forwards each packet onward at some later instant, e_A and e_R respectively. The master therefore observes the interval $e_R - e_A$, up to the propagation and capture delay of the two packets on the link between them.

The interval that carries information about the device is

$$\text{CLRT}_{\text{original}} = t_R - t_A, \quad (1)$$

the time the outstation itself spends between acknowledging the request and producing the answer. This is the cross-layer response time, and it is the feature Formby et al. [10] use, because the work a device does in that window depends on how its vendor implemented the function.

The switch changes what the master sees by holding each packet. We call the two holds

$$D_A = e_A - t_A, \quad D_R = e_R - t_R, \quad (2)$$

the *acknowledgment hold* and the *response hold*. Substituting the two definitions into the interval the master observes gives the identity that governs the whole design,

$$\text{CLRT}_{\text{new}} = e_R - e_A = \text{CLRT}_{\text{original}} + D_R - D_A. \quad (3)$$

Equation (3) makes the difference between two kinds of defense concrete. If the two holds are constants, their difference is a constant, and the observed interval is the original interval moved by that constant. The distribution slides along the axis and keeps its width, so an observer who has seen the device before can still recognize it by the shape. To remove the device's own contribution the holds cannot both be constants: D_R must absorb the variation in $\text{CLRT}_{\text{original}}$.

Our mechanism does this by scheduling both releases from one anchor. When the acknowledgment arrives, the switch computes both release instants immediately,

$$e_A = t_A + D_A, \quad e_R = t_A + D_A + \text{CLRT}_{\text{target}}, \quad (4)$$

where $\text{CLRT}_{\text{target}}$ is the interval we want the master to observe. Because both instants derive from t_A and neither derives from t_R , the term $\text{CLRT}_{\text{original}}$ cancels in (3) and the observed interval is the configured one. The response hold that results is not chosen but implied,

$$D_R = D_A + \text{CLRT}_{\text{target}} - \text{CLRT}_{\text{original}}. \quad (5)$$

Equation (5) is the reason the three quantities cannot be picked independently. Once the acknowledgment hold and the target are fixed, the response hold follows from whatever the device happened to do on that transaction, and it varies from one transaction to the next exactly as much as the device does.

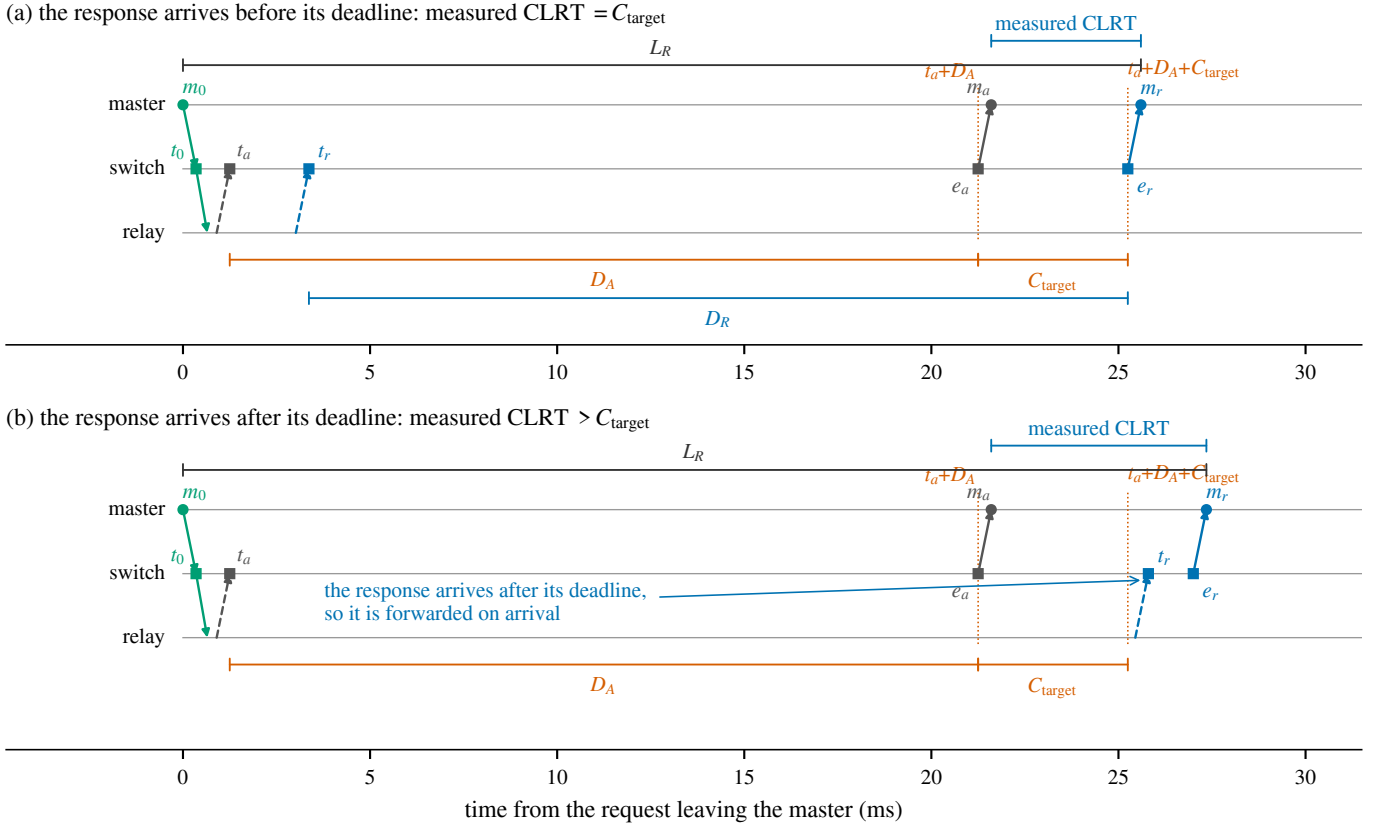


Fig. 3. Release timeline of one READ transaction. The switch arms both deadlines from a single anchor, the arrival of the outstation's acknowledgment at t_a . In (a) the response arrives before its deadline, so the master observes the configured target. In (b) it arrives after, and the switch forwards it on arrival. The response hold D_R is drawn in (a) to show that it is not a configured constant: it is whatever the schedule requires, so it moves with the arrival t_r . Only m_0 , m_a and m_r were measured; the four switch-side instants were not. Propagation and the switch's processing term are drawn at a legible size, not to scale.

This holds only while the response exists in time to be released on schedule. The switch can delay a packet but cannot emit one it has not received, so (4) requires

$$t_R \leq t_A + D_A + \text{CLRT}_{\text{target}}. \quad (6)$$

Note that the acknowledgment may leave before the response arrives; that ordering is normal and does not by itself imply a lost packet. When (6) fails, the deadline is already in the past when the response appears and the switch forwards it immediately. The master then observes an interval larger than the target, as in Figure 3(b). We report these transactions rather than excluding them, because they are the boundary of the mechanism and not a defect of the measurement.

B. Choosing the holds and the target

The model leaves three quantities to decide: the acknowledgment hold D_A , the target $\text{CLRT}_{\text{target}}$, and, through (5), the response hold D_R that follows from them. Each is limited by a different constraint.

The acknowledgment hold is limited in principle by the transport. Holding an acknowledgment delays the feedback the master's TCP uses to decide that a segment is lost, so a hold that approached the master's retransmission timeout would make the master resend a request the outstation had

already answered. Two things make this bound less tight than it first appears. The timer is adaptive: RFC 6298 [15] computes it from the round-trip times the stack has itself observed, so a hold that is applied uniformly is folded into those samples and raises the timeout rather than racing it. The standard notes the same effect from the attacker's side, as a way to inflate a sender's timer [15, §6]. What that argument does not cover is a hold that varies between transactions, which enters the variance term instead. Our policy holds by a constant, which is the case the adaptation absorbs.

In this deployment the transport bound is not the one that binds. The mechanism sustains a hold with a finite resource, described in Section V, and cannot hold a packet past a horizon H that the control plane computes from that resource; for the configuration we evaluate, H is 30.8 ms. Subtracting a conservative bound on the outstation's own acknowledgment latency, the measured detection and release terms, and a safety margin leaves an admissible budget of 24.8 ms, which the control plane enforces before it installs a policy. That ceiling is an order of magnitude below any retransmission timeout the master could plausibly use, so the mechanism reaches its own limit long before the transport notices. We report the margin we observed in Section VI-D and we do not claim the tested hold is universally safe: the master's realized timer was never

read back, and it is the one measurement that would settle the transport question directly.

The target is limited by the operation, and here we could not find a standard that fixes it. We looked for a communications requirement that applies to a DNP3 poll or a select-before-operate control on a distribution relay, and the only numeric deadline we could verify from primary documentation is the relay’s own select-to-operate timeout, whose default is one second. Timing classes defined for other protocols govern those protocols’ own messages and do not transfer to a DNP3 poll merely because the numbers are convenient. We therefore treat the target as a deployment choice rather than a compliance question, state the value we tested, and report the latency the mechanism adds so that an operator with a requirement of their own can check it against that number.

The acknowledgment hold and the target together set what the mechanism costs. Their sum

$$D = D_A + \text{CLRT}_{\text{target}} \quad (7)$$

is the release budget: by (6) it decides which transactions the switch can place on the schedule at all. A larger budget covers more of the device’s own response-time distribution and costs more delay on every transaction it covers. Note that D is the budget, not the delay added to each exchange: a transaction the outstation answers quickly is held longer than one it answers slowly, and the added delay is at most D rather than equal to it.

Because the budget and the observable are different quantities, an operator can fix the cost and still choose the feature. Holding D at 24 ms, we installed targets from 1 ms to 22 ms and the master observed each one; the delay the mechanism added stayed the same across all of them. Section VI-D reports that sweep and the point at which the realized hold stops tracking the requested one.

The framework and the policy it carries are separate. The model above says nothing about what $\text{CLRT}_{\text{target}}$ should be; a constant target is the objective we chose, because a constant removes the device’s variation entirely and is the simplest policy to state and to check. A schedule could instead draw the target from a distribution, or make it a function of the operation, and the same mechanism would carry it. We implement and evaluate the constant target, and we do not claim evidence for the others.

C. The control transaction needs a different anchor

A select-before-operate control command differs in one way that matters here: the framework holds the command itself before it reaches the relay, so that the moment the master issued the command is separated from the moment the relay acts on it. The outstation’s acknowledgment then cannot be the anchor, because that acknowledgment does not exist until after the command has been released, and any deadline measured from it would carry the length of the command hold into the interval the master observes.

The control lane is therefore anchored to the request. When the OPERATE arrives at T_0 , the switch releases it toward the

relay after an internal delay J , and places the acknowledgment and the solicited response the master sees at configured offsets from T_0 ,

$$e_{\text{ack}} = T_0 + A, \quad e_{\text{or}} = T_0 + R. \quad (8)$$

The interval the master observes is then

$$O = e_{\text{or}} - e_{\text{ack}} = R - A, \quad (9)$$

in which J does not appear. Consequently the relay-facing release time can vary while the master-facing interval does not, and an observer cannot subtract a known offset to recover it. The control plane checks that A exceeds the largest admissible J plus the outstation’s own acknowledgment latency before it installs the values, since a deadline the switch cannot honor would fail open rather than protect. In our experiments $R - A$ and $\text{CLRT}_{\text{target}}$ are both 4 ms, so the two lanes present the same value to the observer.

D. What the design does not change

The mechanism replaces one interval per lane and leaves the rest of the exchange as it was. The DNP3 function codes remain in plaintext and the direction of every packet remains visible, so an observer can still tell a poll from a control command by reading the packet. The request-to-acknowledgment interval still contains the outstation’s own acknowledgment latency, displaced by the constant D_A but not reshaped by it, so whatever that interval reveals it continues to reveal. The switch forwards the packets it received without padding them, injecting new ones, or editing any DNP3 field, which is what the plaintext setting requires and which also means the mechanism changes no size or ordering feature. Section VI-F reports what an attacker who is allowed to retrain on the obfuscated traffic still recovers from the features that remain.

V. IMPLEMENTATION

Section IV says when each packet should leave. This section explains how a switching chip can be made to release a packet at a chosen instant, which is not something a forwarding pipeline ordinarily does, and then gives the details needed to judge that the realization is correct and fits the hardware.

A. Holding a packet with queues

The difficulty is that the switch has no general-purpose timer. A P4 pipeline runs to completion for each packet; it can read a clock, but it cannot sleep, and nothing in it can schedule a callback. What the switch does have is a traffic manager with multiple queues per port and a strict-priority scheduler, and a scheduler is a device for deciding what does *not* leave.

We use that. Each lane owns two queues on the same internal port. The *hold queue* carries the packet we intend to delay. The *blocker queue* sits above it in strict priority and carries small packets the switch generates itself. A strict-priority scheduler serves the higher queue whenever it is non-empty, so as long as blockers keep arriving in the blocker

queue, the held packet is not served and stays where it is. The hold ends when the blocker queue runs dry.

The blockers are produced by the on-chip packet generator when the transaction begins, and they circulate on an internal loopback port: they never reach a cable, so neither the master nor the outstation can observe them. Each blocker carries a tag naming its transaction and lane, so a blocker left over from an earlier transaction cannot delay a later one.

B. The deadline decides, the blockers only sustain

It would be natural to read the arrangement above as a timer built from a count: if a blocker takes τ to drain, then K blockers hold a packet for about $K\tau$. That is not how the program measures time, and the distinction matters for what the mechanism guarantees.

When the anchoring packet arrives, the switch stores an absolute release instant in a register, quantized to a 256 ns grid. Every time a blocker recirculates it compares the current timestamp against that stored instant. While the difference is negative the blocker is regenerated and the hold continues; once it is non-negative the deadline has passed, the blocker is not regenerated, the blocker queue empties, and the scheduler serves the held packet. Timing therefore comes from the comparison against a stored deadline, and the blocker population only has to be large enough that the higher-priority queue never runs dry *before* that comparison succeeds. We size the reservoir for continuity, not for duration.

This is what makes the coupling of Section IV-A implementable. The response's release instant is computed as $t_A + D_A + \text{CLRT}_{\text{target}}$ at the moment the acknowledgment arrives, so the response hold that results absorbs whatever the outstation does, without the switch having to measure the outstation at all.

C. One READ transaction end to end

The master's request arrives. The parser extracts the Ethernet, IP, TCP and DNP3 link and application headers, a table classifies the packet by function code as a READ, and the program forwards it toward the relay without delay, recording in a register that a transaction is now in flight together with the acknowledgment number it should expect back.

The relay's transport acknowledgment arrives at t_A . The program recognizes it against the recorded state and writes both release instants at once, $t_A + D_A$ for the acknowledgment and $t_A + D_A + \text{CLRT}_{\text{target}}$ for the response that has not yet arrived. Writing both here, from one timestamp, is what puts them on a common anchor. The write is one-shot from an unarmed sentinel value, so a duplicate acknowledgment cannot arm a second deadline. The acknowledgment itself is placed in the read lane's hold queue and the packet generator is triggered to fill that lane's blocker queue.

The relay's application response arrives at t_R . If its deadline is still in the future the response joins the hold queue behind the acknowledgment. Each deadline expires in turn, its queue drains, and the two packets leave in order at their scheduled instants. If the deadline has already passed when the response

arrives, there is nothing to wait for and the program forwards it immediately; this is the case of Figure 3(b), and it is reported rather than discarded. When the response has left, the program clears the transaction state so the next poll starts from an unarmed register.

D. Keeping the two lanes independent

A read response and a control response can be in flight at the same time. If they shared one hold queue, whichever waited longer would delay the other and the policy the second one was supposed to present would be wrong. Each treatment therefore has its own internal recirculation port, its own hold and blocker queue pair at descending strict priority, and its own register state. Four queues carry the acknowledgment and response holds of the read lane; the control lane's OPERATE path adds its own, and the two sets do not interact.

E. Failing open

A defense on a control path must not become an availability risk, so every path through the program has a forwarding outcome. A packet outside the protected session, a request the program does not recognize, an acknowledgment with no armed transaction, and a response whose transaction has already retired are all forwarded unchanged: such an exchange loses its protection but not its delivery. Every hold is bounded from above by a budget on the number of blocker passes, which releases a held packet and clears the transaction state if its reservoir has not drained within the horizon H of Section IV-B, so a lost acknowledgment or response cannot leave a deadline armed indefinitely. With the timing mode switched off, the same program forwards every packet immediately and arms nothing, which is how we collected the Timing OFF arm.

The program holds and forwards packets and does nothing else to them. It does not pad a response, inject a packet into the protected exchange, or edit a DNP3 field, so every byte the master reassembles is the byte the relay produced and the link-layer checks over those bytes remain valid.

F. The switch program and its resources

We implemented the mechanism as a single P4₁₆ program for the Tofino Native Architecture, compiled with the Barefoot SDE 9.13 compiler, in about 1,560 lines. It occupies all twelve ingress match-action stages of one pipeline and six egress stages, with 112 tables. Transaction state is small: a set of one-entry registers accessed by stateful ALU actions, with one access per packet per register, which is what the architecture permits. The final forwarding decision is one table keyed on a compact outcome computed by the earlier stages, so forwarding, holding, releasing and dropping an expired blocker are each an explicit entry rather than an implicit fallthrough.

One front-panel port at 25 Gb/s carries the master and another at 1 Gb/s carries the relay; these are the only external links. Two further ports are configured in loopback and carry no cable. They are the read and control lanes of Section V-D.

Every policy quantity is a runtime table entry: D_A , the target, the control lane's A , R and admissible J values, the

fail-open budget, and the timing mode. An operator changes the policy by writing table entries, not by loading a different program. The loaded build realizes the dual-deadline schedule only. Its arming table admits a transaction in that mode alone, so the acknowledgment-only and response-only limits of the model are properties of the design rather than settings of this build: when we configured them, the program forwarded both packets unheld and the measured interval was the outstation's own. Reaching them would need a different program, which we did not build and do not claim.

G. Control plane and testbed

The control plane is about 890 lines of Python over the switch's runtime interface. It brings up the ports, creates the queues, installs the multicast and packet-generator configuration, writes the timing parameters, and reads every value back. It also refuses configurations the data plane could not honor: it checks the admissible budget of Section IV-B, and on the control lane it checks that A exceeds the largest admissible J plus a bound on the outstation's acknowledgment latency, and that R exceeds A .

The outstation is a Schweitzer Engineering Laboratories SEL-751A feeder protection relay, reached through an unmanaged switch on the relay-facing port. On the master-facing port a server runs the DNP3 master, which issues READ requests and select-before-operate control operations to isolated control points through a guarded driver, so that no control operation in the evaluation moves a breaker.

VI. EVALUATION

In this section, we evaluate the framework against the five objectives of Section III, RO1 to RO5, on the physical testbed of Section V.

A. What we measured, and where

Evaluation Cases. We compare two arms of the same program. In *Timing OFF*, the switch forwards the outstation's acknowledgment and response as they arrive; in *Obfuscated*, it holds them and releases them on the deadlines of Section IV. The two arms differ only in the release schedule. The switch is the only path between the master and the relay, and, because all timestamps are taken on the master-facing link, every interval reported in this section is one that a passive observer on the control network can record.

Concretely, the interval we report is the time between the arrival of the transport acknowledgment and the arrival of the application response, both observed on the master's own interface. The switch's internal instants, when the outstation's packets reach it and when it releases them, are on links that were not captured, so they are recovered from the program rather than measured.

Dataset. We report per DNP3 exchange, i.e., one request, the transport acknowledgment that follows it, and the application response. The campaign is 22 grouped collection runs in one approximately five-hour window, with six captures of 480 exchanges per run and 132 captures in total. It contains 63,360

exchanges, 31,680 per arm, of which 26,400 are READ, 2,640 SELECT, and 2,640 OPERATE in each arm. Every request was answered, with no retransmission and no malformed frame, and every capture holds 1,448 frames and 130,708 bytes. To characterize the release policy itself, we collected a separate sweep on the same testbed, 5,860 further exchanges over 19 captures: 16 configured release policies of the dual-deadline mode the loaded program realizes, and three control captures taken with the timing mechanism disabled.

B. Does the released interval reach its target? (RO1)

Effectiveness in RO1. We achieve RO1 if the read-path schedule of (4) replaces the outstation's own CLRT with the policy value. We compare the CLRT of READ and of the SELECT phase of SBO between the two arms over all 22 grouped runs as full empirical distributions, because the shape of the Timing OFF distribution is what can make a single observation informative to an adversary.

Figure 5 shows the two READ distributions directly. Before obfuscation the interval is broad and multi-modal, spanning 1.01 to 83.46 ms with distinct peaks near 1.2, 2.1 and 4.0 ms; after it, 99.9% of transactions fall within 0.10 ms of the configured target. The sample variance falls from 6.808 to 0.394 ms², a reduction of 94.2%, and the sample standard deviation from 2.609 to 0.628 ms. The zoom panels show what the full-range view cannot: the released interval is not a single value but a narrow distribution about the target, and the Timing OFF arm places only 13.5% of its transactions anywhere in that window.

In Figure 4(d), we show the median CLRT under Timing OFF is 2.116 ms for READ and 2.050 ms for SELECT, with interquartile ranges of 2.778 ms and 2.697 ms. Under the mechanism, both medians are 4.000 ms and the interquartile range of each falls to 0.006 ms, a reduction of more than two orders of magnitude in spread. Figure 4(a, b) shows why the median alone understates the change: the Timing OFF distributions rise in a small number of discrete steps, so a single observation can be informative about the class, whereas the obfuscated distributions are a step at the scheduled release. This is because the release instant is computed from the outstation's acknowledgment and a fixed offset, and thus, it no longer depends on how long the device took. The change is a replacement, not a shift: the standard deviation falls from 2.609 to 0.628 ms for READ and from 2.267 to 0.024 ms for SELECT, variance ratios of 0.058 (95% CI [0.019, 0.101]) and 0.0001 (95% CI [1.4×10^{-5} , 3.4×10^{-4}]) under a session-level bootstrap. A defense that merely shifted the interval would keep the Timing OFF spread, because translating a sample leaves its variance unchanged: the counterfactual of the Timing OFF distribution translated onto the protected median retains a standard deviation of 2.609 and 2.267 ms, one to four orders of magnitude above what we measure. That counterfactual is analytical, computed from the Timing OFF samples themselves. The loaded program has no shifting mode, so it is not a measured arm and we do not present it as one. The spread collapses because both releases share

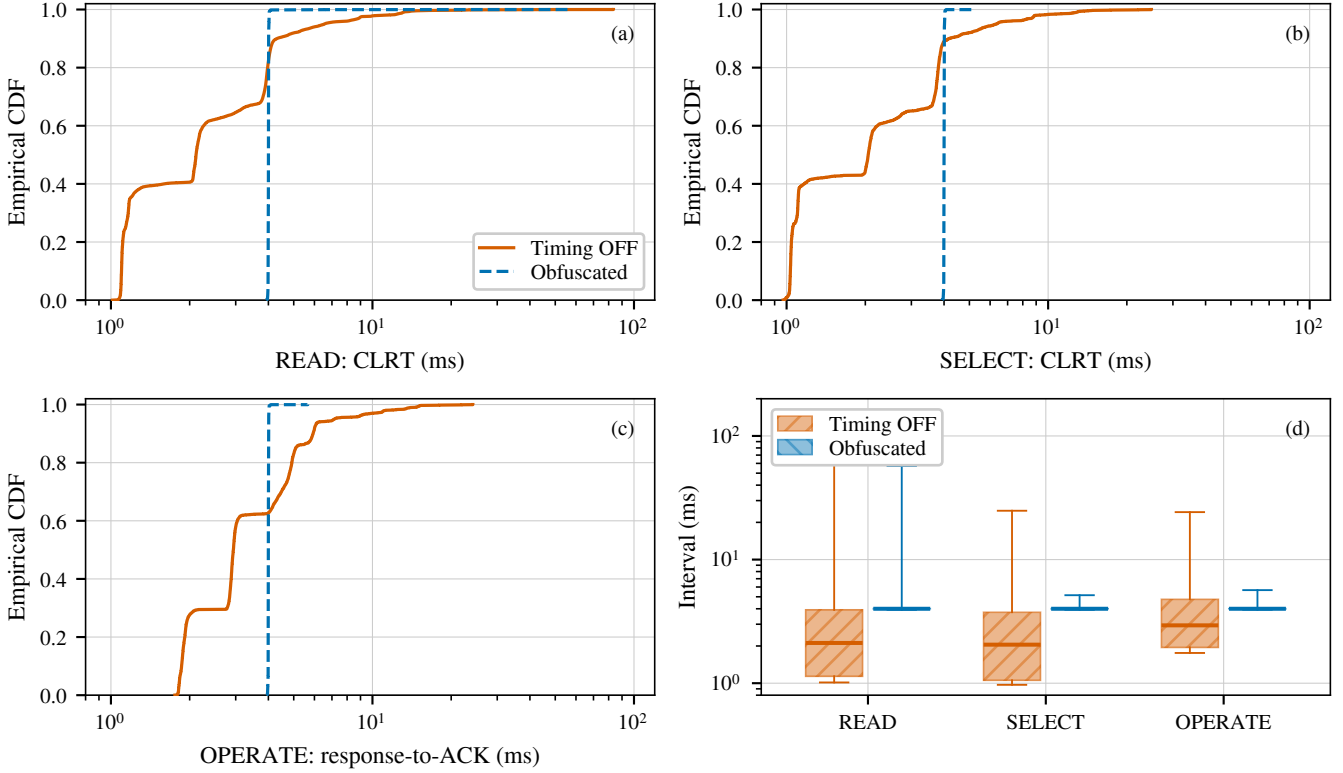


Fig. 4. **Measured interval per transaction class, over 22 grouped runs.** (a) READ and (b) the SELECT phase of SBO report the cross-layer response time; (c) OPERATE reports the master-visible response-to-ACK interval, which is a different anchor and not a complete SBO transaction. The abscissa is logarithmic and spans the full support, so no observation is clipped. (d) The same measurements as quartiles, with whiskers spanning the full support.

the acknowledgment anchor, as Section IV sets out. Figure 6 shows the distinction directly: the analytical reference keeps the Timing OFF shape and merely slides it onto the configured interval, whereas the measured obfuscated distribution concentrates there, and removing each distribution’s own mean leaves that difference intact. Low spread at the wrong interval would not be normalization, so we also report where the distribution sits: both medians are the configured 4.000 ms, and the interquartile range of 0.006 ms is two orders of magnitude below the 1 ms step between neighbouring policy settings in the sweep of Section VI-D. A tail remains, and it is one-sided: the largest obfuscated READ interval is 57.182 ms, and 24 of 26,400 READ exchanges depart from the scheduled release by more than 1 ms, while none falls materially below it. The asymmetry is the mechanism, not the data. A response that reaches the switch after its scheduled release cannot be moved backwards, so it is forwarded on arrival and lands above the target; nothing can place one early. These are the coverage boundary of RO4 seen from the other side.

Stability within the Campaign. In Figure 7, we plot each grouped run in acquisition order. The obfuscated median sits at the scheduled release in every run and for every class, while the Timing OFF medians stay separated. Consequently, the effect is not an artifact of one run or of drift over the five hours.

C. Does the control lane behave the same way? (RO2)

Effectiveness in RO2. We achieve RO2 if the control-path rule of (8) holds the master-visible response-to-ACK interval at the policy value. Because the control path is anchored to the request, its observable is $O = R - A$, in which the internal hold J does not appear, and the read-path budget D does not apply to it. The switch draws J per transaction from the configured codebook $\{2, 6, 12\}$ ms. As shown in Figure 4(c, d), the OPERATE median moves from 2.937 ms under Timing OFF to 4.000 ms under the mechanism, with the interquartile range falling from 2.827 ms to 0.006 ms, and the master-visible interval remained concentrated near the configured 4 ms policy value across all 22 grouped runs. This is a statement about the interval the master sees; the relay-facing release at $T_0 + J$ is not observed (Section VI-G).

D. What range of settings does the mechanism support? (RO4)

Effectiveness in RO4. We achieve RO4 if the interval a passive observer records is a policy value the operator sets, over a usable range of settings. To measure that range on the switch itself, we installed 16 release policies in turn and ran a fixed DNP3 workload through each. Two series answer the question. The first holds the total budget at $D = D_A + \text{CLRT}_{\text{target}} = 24$ ms and moves the split between the two, which changes the visible CLRT while leaving the

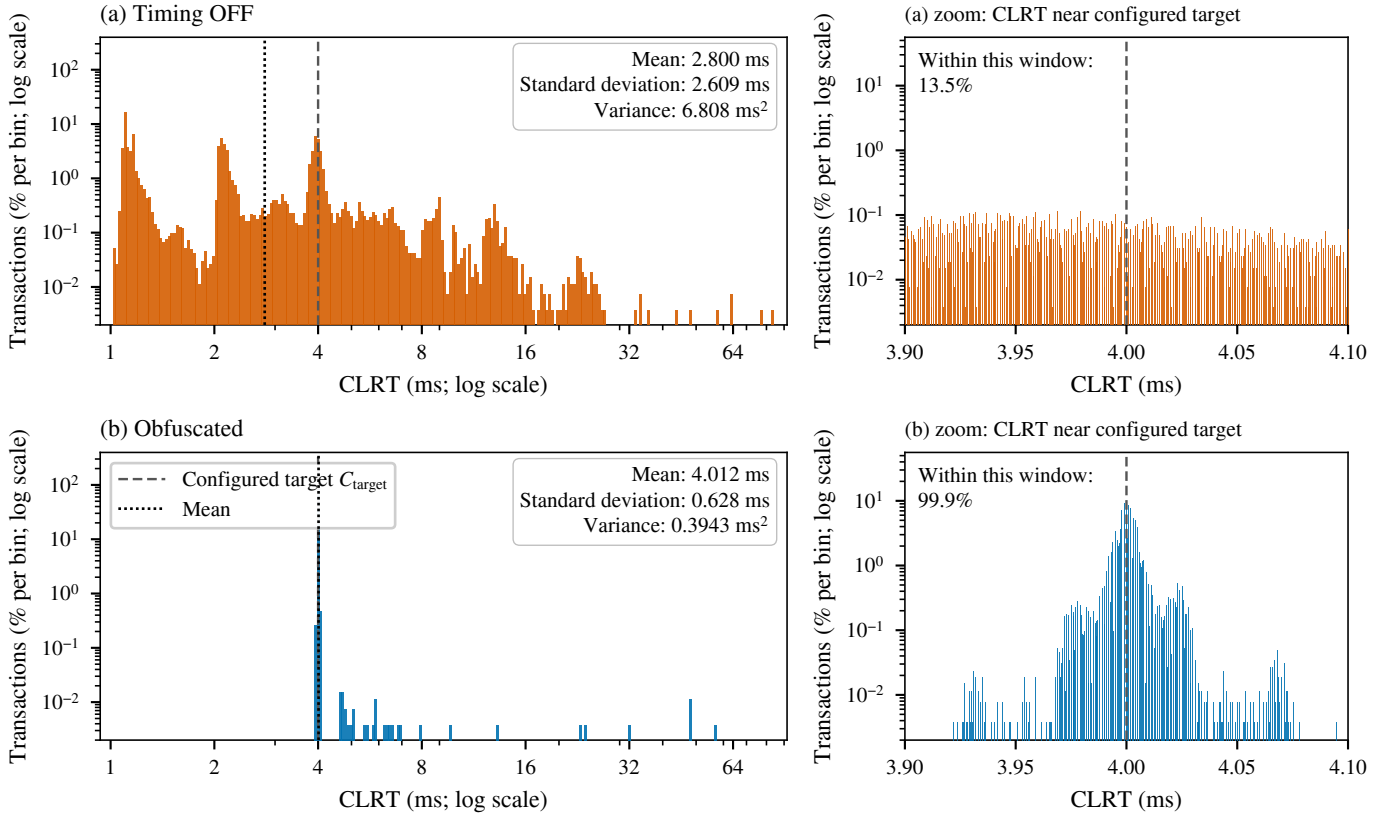


Fig. 5. **Measured READ CLRT before and after obfuscation.** (a) Timing OFF and (b) Obfuscated, over the 22 grouped runs. Left: the full measured range on logarithmic axes, on common bin edges spaced evenly in $\log_{10}$ and identical limits, so a bar in (a) and a bar in (b) mean the same at the same height. Right: a linear zoom on a 0.10 ms window either side of the configured target. Bars are the percentage of that arm’s own transactions in each bin, so the comparison does not depend on sample size; in the zoom the denominator is still the arm’s full sample, and the share falling inside the window is stated on the panel. The dashed line is the configured target C_{target} , a policy value and not a measurement; the dotted line is the measured mean. Nothing is smoothed and no tail is clipped.

total unchanged. The second fixes the target at 4 ms and raises D_A until the mechanism leaves its operating region. Each point is the median over its READ exchanges.

In Figure 8(b), we show that the measured CLRT follows the configured target along the identity line across the whole series: targets of 1, 2, 4, 8, 12, 16, 20 and 22 ms produce measured medians of 0.998, 1.999, 3.999, 8.001, 12.001, 16.000, 20.001 and 22.001 ms, while the end-to-end response time stays between 25.30 and 25.34 ms at every point. As a result, the leaking interval and the cost of the exchange can be set independently, which is what makes the mechanism a framework rather than one fixed setting. In Figure 8(c), the request-to-acknowledgment interval tracks D_A up to 30 ms, where the measured median is 30.571 ms and the CLRT still sits at 4.001 ms. Beyond that point, the acknowledgment interval saturates at about 31.07 ms and the CLRT can no longer be held: it rises to 5.503, 7.498 and 9.507 ms at D_A of 32, 34 and 36 ms. The reason is that the reservoir that starves the queue carries a finite pass budget, so a hold cannot outlast it, and the saturation sits close to the fail-open horizon H of Section IV. From below, the range is closed by the outstation’s own response time, which the switch cannot anticipate: in

Figure 8(a), the budget of 24 ms covers 99.900% of Timing OFF read-lane exchanges, and 29 of 29,040 arrive too late to be held, 28 of them READ and one SELECT.

E. What does the mechanism cost? (RO5)

Effectiveness in RO5. We achieve RO5 if the latency the mechanism costs and the frames and bytes it adds are measured, and the external DNP3 workload still completes. As shown in Figure 8(d), the median response time rises by 22.7 ms for READ, 22.6 ms for SELECT and 21.2 ms for OPERATE. The added latency is close to, but below, the release budget, because it depends on how long the outstation itself took; it is a real operational cost that an operator would weigh against the polling period and any protocol timeout. On the master-facing link, each capture carries 1,448 frames and 130,708 captured bytes for 480 exchanges in both arms, i.e., 3.017 frames and 272.3 bytes per exchange, identical in the two arms. In other words, the mechanism moves packets in time without adding a frame or a byte to that link. Across all 63,360 exchanges, every request was answered with a well-formed response, and all 5,280 SELECT and OPERATE exchanges returned a success status, so the DNP3 workload completes unchanged under the mechanism.

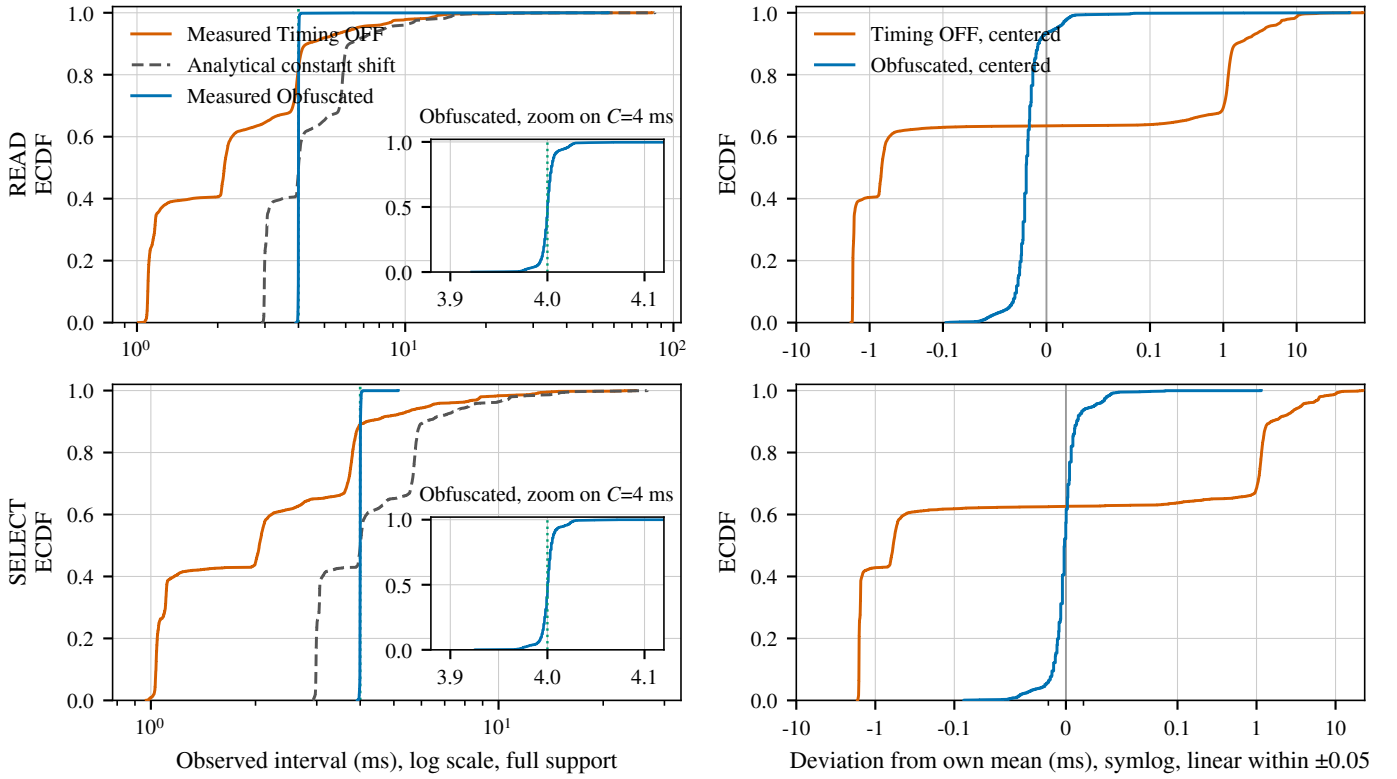


Fig. 6. **A constant translation does not explain the change.** Empirical distributions of the measured CLRT for READ (top) and the SELECT phase of SBO (bottom). Left: the full support on a logarithmic abscissa, so no observation is clipped, with an inset linear zoom on the configured target. Three curves: measured Timing OFF, measured Obfuscated, and an *analytical* constant-shift reference, the Timing OFF sample translated so that its median lands on the configured target, which lands on the configured interval while preserving the Timing OFF variance and shape exactly. That reference is analytical, not an arm: the loaded program has no shifting mode. Right: each measured distribution with its own sample mean removed, which changes neither variance nor shape and so separates variability from location; the analytical reference coincides with the centered Timing OFF curve by construction and is not drawn twice. Late and fail-open observations are included throughout.

Timeout and Retransmission Headroom. Holding the acknowledgment consumes the master’s timers, so we measured what it consumed. On the master-facing captures we recovered every transport-level event across all 63,360 exchanges: the request bytes were acknowledged in a separate segment every time, never piggybacked on the response, and no request was sent while earlier bytes were still unacknowledged. There was no retransmission, no reset and no duplicate acknowledgment in either arm. The longest the master waited for its request to be acknowledged was 29.2 ms and the longest it waited for the response was 77.7 ms. We did not record the master’s kernel configuration, so its actual retransmission timeout is unknown; for scale, the minimum Linux documents is 200 ms and RFC 6298 recommends a 1 s floor, and no retransmission occurred. The receive timeout in our driver is 3 s, and it bounds one read rather than the whole transaction, so it is not a transaction deadline; we report the measured completion times against it without asserting that the two coincide, which would need application-level receive tracing we did not collect. Select validity is the other timer a held control transaction could violate, since the master issues its OPERATE only 0.19 ms after the SELECT response reaches it and the mechanism delays that response: none of the 2,640 obfuscated

OPERATE exchanges was refused for a stale select. All of this characterises this operating point on this testbed, and would have to be rechecked for a master with a shorter timeout.

F. What does the attacker still learn? (RO3)

Setup. We achieve RO3 if the timing features carry less information about the transaction class and a fixed adversary trained on undefended traffic is reduced to chance. We treat this as a three-class problem over READ, SELECT and OPERATE, for which chance balanced accuracy is one third, with the two independent intervals, request to acknowledgment and acknowledgment to response, as features; the total response time is their sum and is excluded. We estimate mutual information for the CLRT in bits and compare it against a null built by shuffling class labels within each grouped run over 1,000 permutations, which preserves the per-run class counts. Every fold leaves out one grouped run, and we report the spread of the 22 held-out-run scores as a range.

Effectiveness in RO3. In Figure 9, we show what the mechanism does to the two measurable intervals before any classifier is applied. Under Timing OFF, the three classes are separated mainly along the post-acknowledgment axis, with median intervals of 2.116, 2.050 and 2.937 ms for READ,

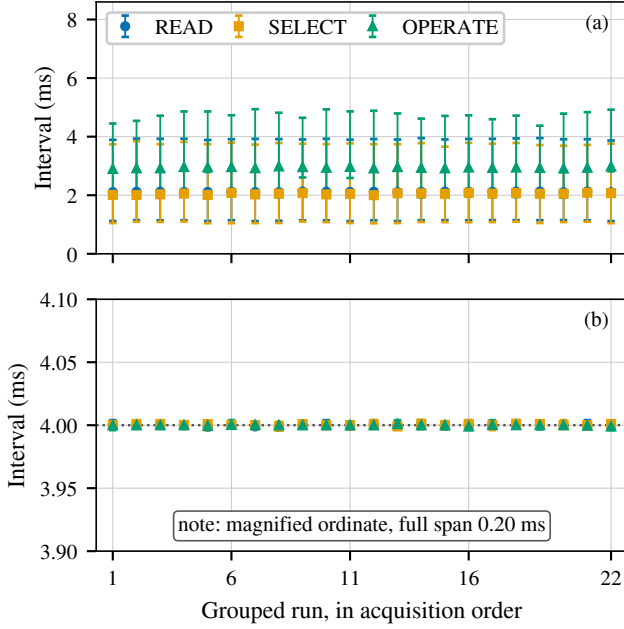


Fig. 7. **Within-campaign stability across the 22 grouped runs**, in acquisition order. (a) Timing OFF and (b) Obfuscated. Markers are the run median and bars span the interquartile range. Panel (b) uses a magnified ordinate spanning only 0.20 ms around the scheduled release. The 22 runs come from one approximately five-hour campaign on one relay behind one switch and are not independent replications.

SELECT and OPERATE against request-to-acknowledgment medians of 0.555, 0.555 and 0.558 ms, which barely separate the classes at all. Under the mechanism, that axis collapses: all three medians sit at 4.000 ms, and READ and SELECT overlap. What remains is a horizontal offset, with the request-to-acknowledgment median moving to 21.336, 21.239 and 20.650 ms. This is because the read lane is anchored to the outstation’s acknowledgment and the control lane to the request.

In Figure 10, we show the two attacker models. The fixed adversary trained on Timing OFF traffic reaches 0.651 balanced accuracy on held-out Timing OFF exchanges using the CLRT alone, and 0.733 using both intervals. Applied unchanged to obfuscated traffic, it falls to 0.333 and 0.334, which is chance. Mutual information between the CLRT and the class falls from 0.383 bits to 0.004 bits; the Timing OFF value lies far above its permutation null at the resolution of the test ($p = 0.001$ with 1,000 permutations), and the obfuscated value lies inside its null ($p = 0.096$). For the evaluated fixed Random-Forest attacker, then, three-class transaction identification falls to approximately chance, and the cross-layer response time stops carrying information about the class. The adaptive adversary, retrained on obfuscated traffic, remains at chance on the CLRT alone but recovers to 0.651 when the acknowledgment latency is added. The reason is that the read and control paths are anchored differently, so their acknowledgment latencies differ, and an adversary willing to

collect obfuscated traffic and retrain can exploit that difference; the confusion panels show its shape, with OPERATE separable while SELECT stays confused with READ. Closing this residual requires the two paths to share an anchor, which we leave to future work.

G. Limitations

One Device, One Campaign. The evidence is one SEL-751A behind one Tofino-1, in one approximately five-hour campaign of 22 grouped runs. Consequently, it does not establish behavior across days, devices, or deployments, and the runs are grouped collections rather than independent replications.

What the Classifier Shows. DNP3 function codes remain visible in plaintext, so the result concerns the timing channel and not the concealment of operations. The classification we evaluate separates transaction classes on one outstation; it is not device-model identification, and with one outstation nothing here shows that two devices become indistinguishable. The attacker results characterize the Random-Forest attacker we evaluated rather than every classifier, and an adaptive adversary recovers part of the class signal from the difference between the two anchors.

What Was Not Observed. Only the master-facing link was captured. Consequently, the realized per-transaction hold J , the relay-facing release at $T_0 + J$, and any physical actuation are configured rather than observed, we report no result per J , and nothing here shows that exactly one OPERATE reached the relay. The fail-open horizon H is computed in the control plane rather than enforced by the data plane, so the saturation of RO4 is an observed transition and not a proof that the two coincide. The sweep offsets come from the configuration table rather than from a per-point readback. The switch’s own release instants were likewise not captured, and the loaded program cannot supply them: the timestamp registers it declares are never written, and every write it does declare would take an ingress timestamp rather than an egress one, so obtaining a release instant would require a modified program. And because the program suppresses a retransmission of the response that matches an already-seen transport position, a retransmission by the outstation of a held response would be absorbed inside the switch and would not appear in a master-facing capture, so we cannot exclude one.

Scope. A fixed read-path release budget leaves a measurable late tail rather than eliminating it. The overhead of RO5 counts the master-facing link; the switch’s internal blocker traffic never appears there. Size obfuscation is outside this paper: the mechanism changes no packet size, and we make no size, padding, splitting or segmentation claim.

VII. RELATED WORK

Device fingerprinting. Devices can be identified from what they emit, even without any access to the payload. Kohno et al. fingerprint a remote host from the skew of its clock as seen in TCP timestamps [16], and GTID fingerprints a device and its type from the timing signature of its packets [17].

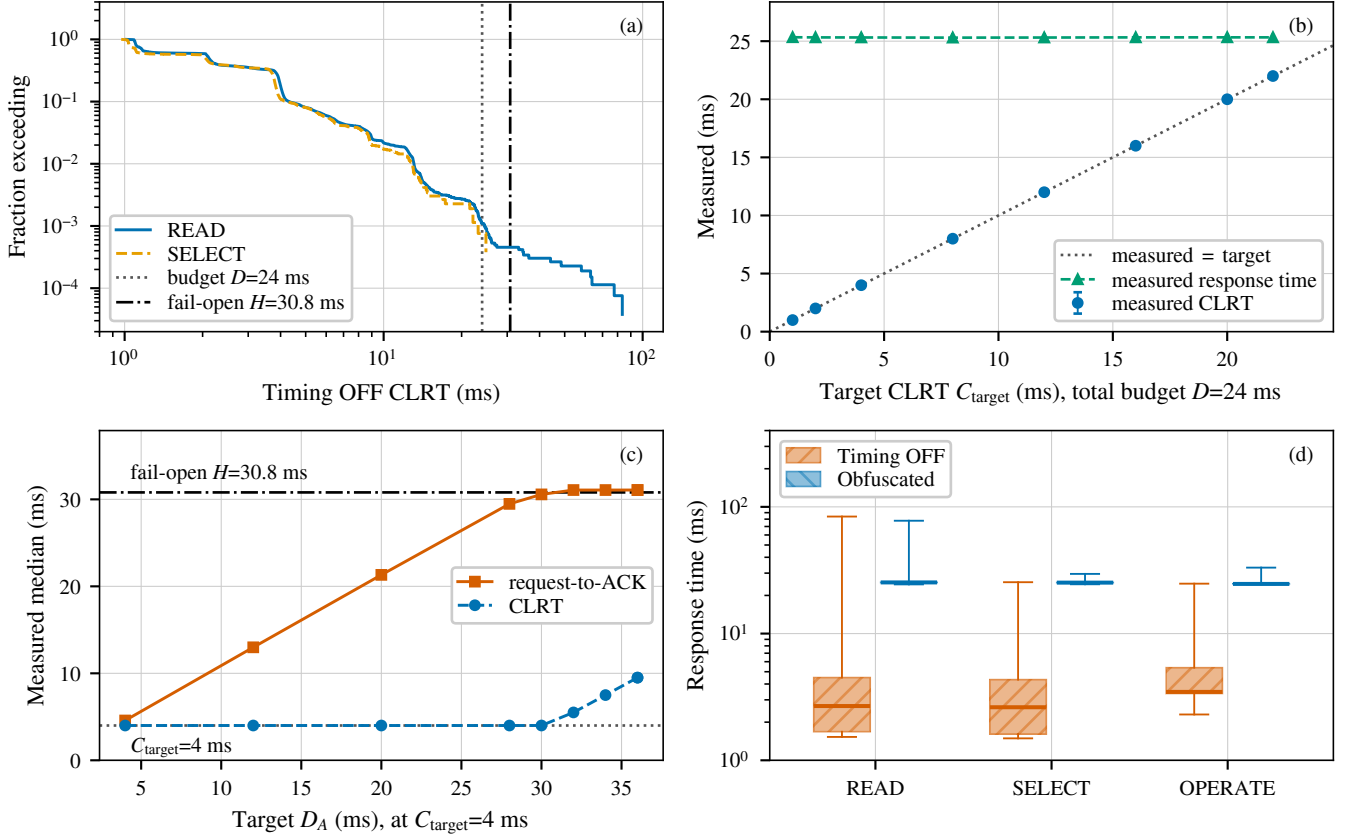


Fig. 8. **The release policy is programmable, and its budget is bounded on both sides.** (a) Fraction of Timing OFF read-lane exchanges, READ and the SELECT phase of SBO only, whose CLRT exceeds a given value, with the release budget D and the fail-open horizon H . (b) Measured hardware sweep at a fixed total budget $D = 24$ ms: the visible CLRT follows the configured target along the identity line while the end-to-end response time stays put. (c) Measured sweep of D_A at a fixed target: the request-to-acknowledgment interval tracks the target until it saturates near H , beyond which the CLRT can no longer be held at its target. (d) End-to-end response time per class. Bars in (b) span the interquartile range; boxes in (d) span the quartiles with whiskers over the full support.

These works define the threat we address; our objective, on the other hand, is to remove the timing features they rely on from the master-facing observation. Their target is the device type, whereas the classifier we evaluate separates transaction classes on one outstation, so we suppress a feature their cross-layer response-time method uses without demonstrating that two devices become indistinguishable.

ICS and power-system fingerprinting. Formby et al. brought fingerprinting to control systems with the cross-layer response time and the physical operation time, measured at the device, of DNP3 outstations [10], and Gu et al. added device physics as a feature [18]. Jeon et al. fingerprint SCADA devices passively without deep packet inspection [19]. The reconnaissance these techniques serve enables false-data injection and process-control attacks [14], [20]. Defenses in this setting have so far worked at the content and connectivity layer: DefRec virtualizes physical functions to present deceptive content [21], RAINCOAT randomizes data acquisition and injects decoy measurements [22], and a companion testbed evaluates such anti-reconnaissance approaches on grid infrastructure [23]. Compared to these, our framework works below

the semantics they reshape and complements them: it changes when the wire carries the outstation’s answer, not what the answer says.

General traffic obfuscation. Traffic morphing transforms one class’s packet-size distribution into another’s [3], while Dyer et al. show that per-packet padding and fragmentation still leave exploitable features [4]. Tamaraw fixes packet size and rate at a bandwidth cost [6], and WTF-PAD pads inter-packet gaps adaptively [24]. However, deep fingerprinting attacks defeat several of these [5], and their accuracy at Internet scale is contested [25]. The same size-and-timing channel identifies smart-home devices under encryption, where shaping is the countermeasure [26], and Pacer and NetShaper shape a tenant’s traffic to a schedule that bounds side-channel leakage in the cloud [27], [28]. These defenses assume an encrypted payload and a cooperating end host or hypervisor that can add cover traffic. Our framework, in contrast, targets a plaintext protocol whose feature is one interval inside a request-response exchange, adds no byte, and runs where the outstation cannot be changed.

Programmable data-plane traffic transformation. P4

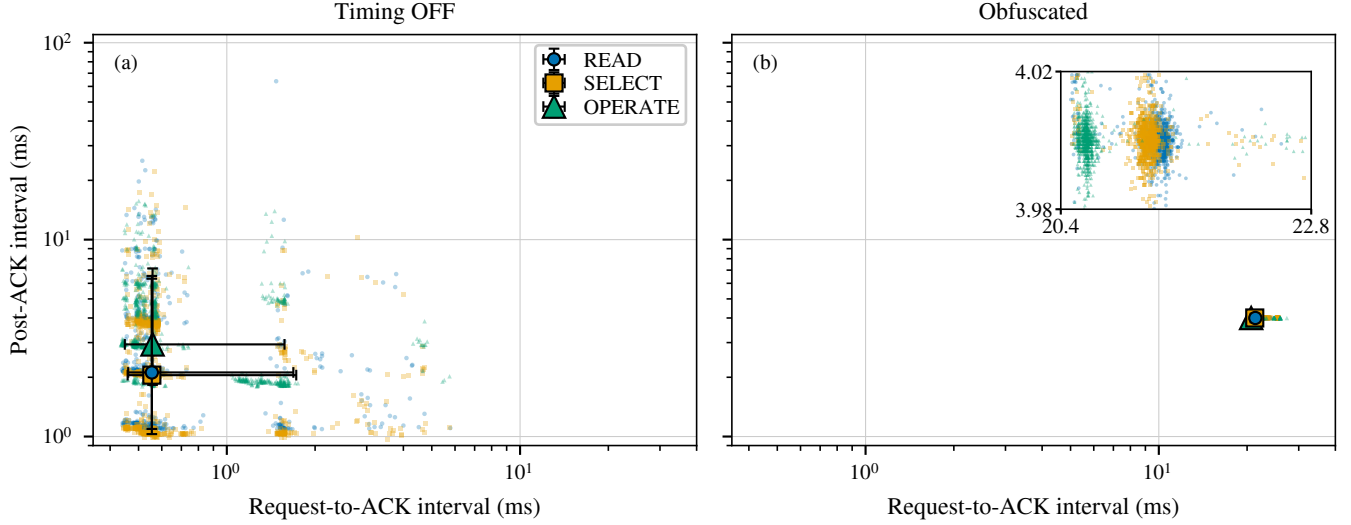


Fig. 9. **Targeted timing-feature collapse and residual anchor leakage.** The two intervals a passive observer can measure, plotted against each other on identical logarithmic axes: (a) Timing OFF and (b) Obfuscated. The ordinate is the post-acknowledgment interval, the cross-layer response time for READ and the SELECT phase of SBO and the master-visible response-to-ACK interval for OPERATE. Small marks are a deterministic, class-stratified subsample drawn for legibility; the large markers are the median and the bars the 5th to 95th percentile, both computed on the complete dataset. Under the mechanism the vertical, device-derived interval of all three classes collapses onto the policy value and READ and SELECT overlap, while OPERATE keeps a horizontal offset because the control lane is anchored to the request and the read lane to the outstation’s acknowledgment. The inset magnifies the collapsed band on linear axes. This figure shows timing-feature overlap among transaction classes on one physical outstation. It is not clustering performance, not device identification, and not evidence that different devices become indistinguishable.

defines the programmable parser and match-action model we use [12]. PIFO defines programmable scheduling at line rate [29], and SP-PIFO approximates it with strict-priority queues on today’s switches [30], which is the primitive our reservoirs use. Ditto pads and injects chaff to a fixed pattern at line rate [7], Minos morphs and schedules encrypted traffic on a programmable switch with low overhead [8], and Securitas mixes fragmentation and insertion under a learned policy across several data planes [9]. NetHide obfuscates topology in the data plane against link-flooding reconnaissance [31]. Closest to our setting, Hu et al. use programmable switches to enhance industrial-protocol security [32]. Our framework differs from these in the observable it addresses, i.e., the master-visible timing of a DNP3 transaction, and in holding packets rather than adding or reshaping them.

DNP3 timing and security. East et al. catalogue attacks on DNP3 and the absence of encryption that makes its semantics visible [13], and specification-based and state-based intrusion detection has been adapted to DNP3 [33], [34]. These do not change the timing an observer sees. On the control path we report a different quantity from theirs: the master-visible OPERATE response-to-acknowledgment interval, which is a protocol-response timing feature. Formby et al. estimate physical operation time from a sequence-of-events-recorder timestamp carried in the outstation’s application payload, and report that the alternative estimate from unsolicited-response arrival times produced no usable results [10]. A mechanism that moves packets in time and changes no byte cannot alter such a payload timestamp, and the traffic we evaluate carries

neither unsolicited responses nor any time-bearing object. We therefore make no claim about physical operation time, and we do not present our control-path result as a correlate of theirs.

VIII. CONCLUSION

Response timing lets a passive observer on a DNP3 control network learn what an outstation is doing, and the endpoints that leak it cannot be modified. We presented a timing-obfuscation framework that runs in the data plane of one programmable switch, holds the packets that carry an outstation’s timing, and releases them on operator-set offsets, with one rule anchored to the outstation’s acknowledgment for polls and the SELECT phase of control, and another anchored to the request for OPERATE. Implemented as a P4 program on an Intel Tofino-1 and evaluated against a physical SEL-751A relay over 63,360 DNP3 exchanges, it moved the cross-layer response time to the configured release value and reduced its interquartile range from about 2.8 ms to 0.006 ms, and a measured sweep on the switch showed the observable following the configured offset across a usable range while the cost of the exchange stayed fixed, and no retransmission or timeout was observed under it. For the fixed Random-Forest attacker we evaluated, one that trains on Timing OFF transaction-class timing before deployment and then applies that model unchanged, three-class transaction identification fell from 0.651 balanced accuracy to approximately chance. The result is bounded: it covers the exchanges the release budget can schedule, it leaves the acknowledgment latency of the two paths distinguishable to an adversary willing to retrain,

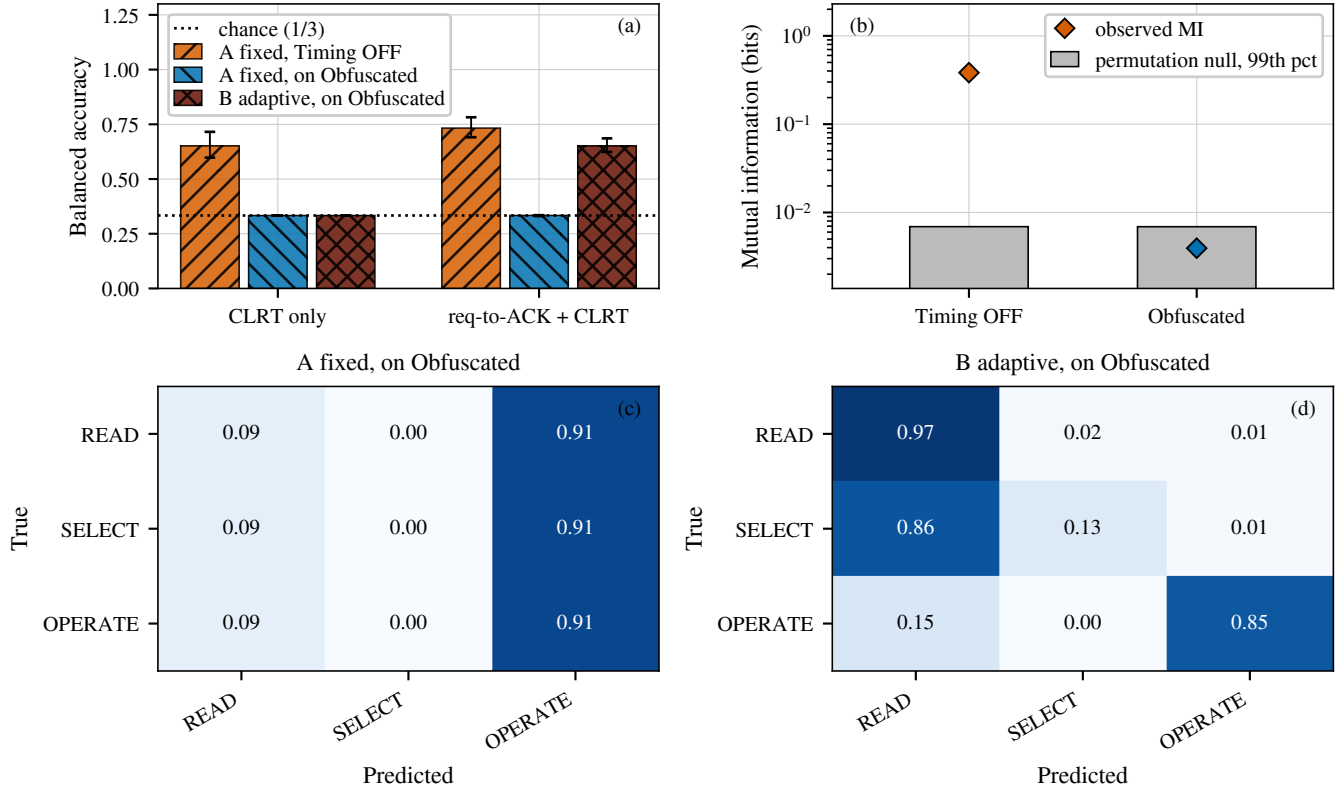


Fig. 10. **Transaction-class leakage under the two attacker models**, leaving out one grouped run at a time over all 22 runs. (a) Balanced accuracy of the evaluated Random-Forest attacker; bars are the mean over the 22 held-out runs and the whiskers span the full range across those runs, which is within-campaign variability and not a confidence interval, because the folds share training data. (b) Observed mutual information between the CLRT and the transaction class against the 99th percentile of a within-run permutation null. (c) and (d) Row-normalised confusion over all 22 held-out runs, using both intervals.

it characterises the attacker we evaluated rather than every classifier, and it concerns one relay behind one switch over one campaign. The plaintext function codes still name each operation, so what the mechanism removes is the additional timing signature of that operation, and with one outstation on the testbed this is the suppression of a class-conditioned timing feature rather than a demonstration of resistance to device identification. Extending the evaluation to several outstation models and to campaigns spanning days would show whether a single release policy holds across devices and over time.

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